Infertility subtype and reference population composition shape China's contrasts with other G20 countries: an ecological analysis of GBD 2023

Author names, institutional addresses and corresponding author
pending research-team completion

25 October 2026

Abstract

Background: International comparisons of infertility can conceal differences between primary and secondary infertility and changes in comparison-population composition. We assessed these issues for infertility attributed to polycystic ovarian syndrome (PCOS) and endometriosis in China.

Methods: This ecological secondary analysis used Global Burden of Disease 2023 estimates for females aged 15–49 years, 1990–2023. China was compared with the other 18 sovereign G20 countries, pooling cases and populations within age groups before standardization. We partitioned changes in the standardized China-minus-reference difference by infertility subtype and, in a post hoc symmetric decomposition, by Chinese rates, reference-country rates and reference-country population shares. Sensitivities varied countries, ages and terminal years. Derived estimates lacked propagated uncertainty because aligned draws were unavailable.

Results: China's PCOS-related infertility standardized prevalence increased from 333.56 to 605.08 per 100,000, versus 606.48 to 726.45 in the reference. The negative difference narrowed by 151.54 per 100,000. Chinese rate changes contributed +271.51, reference-country rate changes −208.62, and reference composition +88.65 (58.5% of the net change). Secondary infertility accounted for 69.4% of the total difference change; this was 69.0–70.5% when ending in 2021 or 2022, and 78.1% with fixed 1990 reference-country shares in 2023. The 2023 primary PCOS difference remained negative under all country exclusions; the small secondary difference (+2.16) changed sign. Endometriosis primary (−7.94) and secondary (+7.73) differences nearly canceled in 2023, but the total difference was −8.76 in

2022 versus −0.20 in 2023.

Conclusions: Apparent convergence in PCOS-related infertility partly reflected changing reference composition and concealed contrasting subtypes. Endometriosis near parity was endpoint-sensitive. These modeled, descriptive contrasts do not establish reproductive-health improvement or causal explanations.

Plain English summary

Infertility estimates are often combined into one number. This can hide differences between people who have never had a live birth and those who have had a live birth previously. Comparisons between countries can also change when some countries make up a larger share of the comparison population.

We studied estimates of infertility linked to polycystic ovarian syndrome and endometriosis among women aged 15 to 49 in China and 18 other G20 countries. China's estimate for infertility linked to polycystic ovarian syndrome increased and moved closer to the comparison estimate. Part of this movement arose because the population shares of the comparison countries changed. The combined result also concealed different patterns for the two types of infertility. For endometriosis, conclusions based on the final year differed from those based on the preceding years.

These findings show why international comparisons need more detail than a single combined number. They do not tell us whether an individual woman's chance of pregnancy changed or whether healthcare improved. The estimates come from statistical models, and we could not calculate the full uncertainty of our comparisons.

Keywords: infertility; polycystic ovarian syndrome; endometriosis; GBD 2023; primary infertility; secondary infertility; population composition; China

1 Background

Infertility associated with polycystic ovarian syndrome (PCOS) and endometriosis is relevant to population reproductive health as well as individualized clinical care [1, 2]. Estimates attributed to these causes within the Global Burden of Disease (GBD) infertility impairment describe a different outcome from total PCOS or endometriosis prevalence. They are also distinct

from lifetime infertility or infertility measured exclusively among people currently trying to conceive [3].

Much of the descriptive territory has already been covered. Xu and colleagues analyzed both underlying diseases and their attribution to total, primary and secondary infertility in GBD 2021 [4]. Lu and colleagues examined the same two infertility causes among women aged 15–49 in China and globally, including projections [5]. Wang and colleagues subsequently analyzed infertility causes and primary/secondary attribution using GBD 2021 and 2023 [6]. Broad trend, decomposition and prediction analyses have also appeared in this journal [7]. Neither a newer GBD edition, subtype stratification alone, nor adding a G20 comparison constitutes a sufficient methodological advance.

A narrower unresolved question concerns the interpretation of an aggregate international contrast. A population-pooled reference combines country rates with country population shares that may change over time, even after age standardization. A narrowing difference could therefore reflect changing Chinese rates, changing reference-country rates, or changing reference-country composition. Primary and secondary infertility may contribute differently to each component. These are separable accounting questions; they do not identify causal mechanisms.

We compared China's cause-specific infertility estimates with a fixed set of 18 other sovereign G20 countries. We quantified subtype and age contributions to changes in the contrast and added an explicit decomposition of reference composition during manuscript review. We also assessed country and endpoint dependence. Endometriosis was retained as a contrasting cause, with terminal-year instability treated as a finding requiring caution rather than evidence of sustained convergence.

2 Methods

2.1 Design, data and outcome definitions

This ecological secondary analysis used GBD 2023 Results Tool downloads obtained on 25 October 2026 [8]. The analysis included females aged 15–49, seven five-year age groups and annual estimates for 1990–2023. China was compared with Argentina, Australia, Brazil, Canada, France, Germany, India, Indonesia, Italy, Japan, Republic of Korea, Mexico, Russian Federation, Saudi Arabia, South Africa, Türkiye, United Kingdom and United States of America. These

are the other sovereign country members of the G20; regional organizations were excluded. The G20 is an economically defined comparison, not a matched control population or a global benchmark.

We selected infertility impairment 198, underlying causes endometriosis (607) and PCOS (605), prevalence and years lived with disability (YLDs), and Number and Rate. Primary (220) and secondary (221) infertility were downloaded separately. Matched female populations came from the same release. Main analyses concern prevalence; YLDs, case-count decompositions, country distributions, trend models and failed forecast validation are secondary results in Additional file 1. No individual records were accessed.

The GBD methods appendix describes survey-based primary infertility in couples without a previous live birth who desire a child, have been in a union for at least five years, and report no contraceptive use. Secondary infertility also uses the interval since the last live birth. Heterogeneous survey and clinical definitions are harmonized within GBD; the resulting estimates should not be equated with a uniform 12-month clinical definition. Underlying-cause allocation occurs within the modeled infertility envelope [3]. The denominator here is all females in each age group, not only couples attempting pregnancy.

2.2 Validation and standardization

We required complete dimensional coverage, unique cells, consistent cause and infertility identifiers, finite non-negative values and ordered original marginal uncertainty bounds. Number divided by matching population was compared with official Rate. Annual subtype rates were reconstructed from Number and matched population; independent original Number/Rate exports for 1990 and 2023 checked this reconstruction. Primary plus secondary point estimates were required to reproduce the total. Input hashes, mappings and retrieval specifications accompany the code [9].

We used published GBD 2023 standard-population weights reproduced to six decimals, normalized over ages 15–49 [10]. The original machine-precision IHME weight file was not retrieved. Thus, all standardized rates are restricted-age rates. With normalized age weight w_a, country c, age a, year t, infertility type k, modeled cases C and female population P, rates per 100,000

were $r_{catk} = 10^5 C_{catk} / P_{cat}$. The age-specific reference was formed before standardization:

$$r_{Ratk} = \sum_{c \in R} s_{cat} r_{catk}, \quad s_{cat} = \frac{P_{cat}}{\sum_{j \in R} P_{jat}}, \quad A_{Ctk} = \sum_a w_a r_{Catk}, \quad A_{Rtk} = \sum_a w_a r_{Ratk}. \tag{1}$$

Here R always excludes China. The country-share weights s vary within age and year; the standard age weights w are fixed. Rates were not constructed by averaging country standardized rates.

2.3 Subtype and reference-composition accounting

We calculated the signed difference $D_{tk} = A_{Ctk} - A_{Rtk}$ and ratio A_{Ctk} / A_{Rtk}. Because primary and secondary counts sum to total, their differences and changes also sum exactly. Their signed contributions are not attributable fractions or causal effects. Each age group's contribution is its standard weight multiplied by its age-specific difference change.

During manuscript review, we added a symmetric decomposition between start year 0 and end year T:

$$\Delta D_k = X_k + Q_k + S_k, \tag{2}$$

$$X_k = \sum_a w_a (r_{CaTk} - r_{Ca0k}), \tag{3}$$

$$Q_k = -\sum_a w_a \sum_{c \in R} \frac{s_{ca0} + s_{caT}}{2} (r_{caTk} - r_{ca0k}), \tag{4}$$

$$S_k = -\sum_a w_a \sum_{c \in R} \frac{r_{ca0k} + r_{caTk}}{2} (s_{caT} - s_{ca0}). \tag{5}$$

X is the Chinese rate component, Q the reference-country rate component and S the reference country-share component. The mean of the two replacement orders shares the product interaction symmetrically. Components sum exactly to the change in the standardized contrast, separately by subtype and age. A positive component moves China minus reference upward; with an initially negative contrast, it can indicate narrowing without indicating health improvement. This decomposition describes reference weighting, whereas the separate case-count decomposition concerns China's own population size and age distribution.

2.4 Sensitivity analyses and uncertainty

We excluded each reference country in turn, restricted ages to 20–49, and fixed within-age reference-country shares at their 1990 values. The symmetric decomposition was repeated for country exclusions, the restricted age range, periods starting in 1990 or 2010, and endpoints 2021, 2022 and 2023. Endpoint analyses were added after inspection of a terminal-year endometriosis increase; the reference-composition decomposition was added during manuscript review. Neither was externally preregistered. Fixed-share contrasts are a separate, order-specific sensitivity, not an interchangeable estimate of the symmetric share component.

We examined all 128 combinations of perturbing each unnormalized standard weight by half of its last published decimal. These deterministic rounding envelopes do not measure epidemiological uncertainty. Additional file 1 describes all secondary methods, including conditional trend intervals and forecast-validation rules.

Jointly aligned GBD draws across ages, countries and years were unavailable. We therefore report derived point estimates without propagated uncertainty intervals or significance tests. Original marginal limits are preserved but not summed or treated as standard errors. Country exclusions describe reference sensitivity, not sampling uncertainty. YLDs and prevalence share inputs and are not independent replications. Python 3.12 and pinned NumPy, pandas, SciPy, statsmodels and Matplotlib dependencies were used. Code and machine-readable outputs are available in Additional file 2 and the repository [9]. Applicable GATHER items, including incomplete propagated uncertainty reporting, are mapped in Additional file 1 [11].

2.5 Use of artificial intelligence

OpenAI Codex was used for coding, execution assistance, source retrieval and manuscript drafting, including this revision. No artificial observations were substituted for the downloaded GBD data. Final human review of code, sources, interpretation and all author declarations remains necessary before submission. The tool is not an author and cannot approve the manuscript.

3 Results

3.1 Coverage and overall contrasts

The total burden export contained 36,176 rows and the population export 4,522 age–country–year cells. A further 36,176 annual subtype Number rows and 4,256 endpoint Number/Rate rows covered both causes and measures. Primary plus secondary reproduced total point estimates throughout. Reconstructed subtype endpoint rates agreed with original rates to within 1.51×10^{-11} per 100,000. These checks establish extraction and arithmetic consistency, not independent validity of GBD estimates.

China's PCOS-related infertility standardized prevalence increased from 333.56 to 605.08 per 100,000, compared with 606.48 to 726.45 in the reference (Table 1; Figure 1). The difference changed from -272.92 to -121.38 and the ratio from 0.550 to 0.833. For endometriosis, rates changed from 63.70 to 53.30 in China and 66.63 to 53.51 in the reference. The 2023 total difference was -0.20; it was -8.76 in 2022. China's endometriosis estimate rose by 23.6% in the last year, so the terminal comparison does not summarize a stable trajectory.

Table 1: Total infertility prevalence in China and the reference population.

		Cases, thousands		Rate per 100,000			
Cause	Year	China	Reference	China	Reference	*D*	Ratio
Endometriosis	1990	192.8	375.6	63.70	66.63	-2.93	0.956
Endometriosis	2023	181.6	457.1	53.30	53.51	-0.20	0.996
PCOS	1990	1083.3	3446.2	333.56	606.48	-272.92	0.550
PCOS	2023	1866.7	6124.2	605.08	726.45	-121.38	0.833

Rates standardized within ages 15–49. Reference pools the other 18 sovereign G20 countries within age. *D*: China minus reference. All cells are modeled point estimates without propagated uncertainty.

3.2 Primary and secondary infertility

In 2023, the PCOS primary-infertility difference was -123.54 per 100,000 (ratio 0.441), whereas the secondary difference was +2.16 (ratio 1.004; Table 2; Figure 2). Their changes since 1990 were +46.40 and +105.14, respectively, summing to +151.54. Thus secondary infertility accounted for 69.4% of the signed total change. The share was 70.5% ending in 2021 and 69.0% ending in 2022. Across country exclusions and restriction to ages 20–49, the 2023 share ranged from 64.2% to 75.4%; fixing reference-country shares at 1990 yielded 78.1%. The exact per-

centage therefore depends on reference construction, while secondary infertility remained the larger positive component in these checks.

The 2023 PCOS primary difference remained negative under every country exclusion (−201.52 to −101.89), but the secondary difference ranged from −113.63 to +30.72. Near parity of secondary infertility was therefore a main-reference result, not a stable sign conclusion. For endometriosis, the 2023 primary difference of −7.94 and secondary difference of +7.73 nearly canceled. Both subtype signs survived country exclusions in 2023, but the secondary difference was −0.40 in 2022. Its endpoint dependence limits the interpretation of cancellation.

Table 2: Primary and secondary infertility prevalence at the endpoints.

Cause	Type	Year	China rate	Ref. rate	*D*	Ratio
Endometriosis	Primary	1990	4.92	19.64	-14.72	0.250
Endometriosis	Primary	2023	4.94	12.87	-7.94	0.384
Endometriosis	Secondary	1990	58.78	46.98	11.79	1.251
Endometriosis	Secondary	2023	48.37	40.63	7.73	1.190
PCOS	Primary	1990	44.70	214.64	-169.94	0.208
PCOS	Primary	2023	97.35	220.89	-123.54	0.441
PCOS	Secondary	1990	288.86	391.84	-102.98	0.737
PCOS	Secondary	2023	507.72	505.56	2.16	1.004

Rate units: per 100,000 females; standardization restricted to ages 15–49. Point estimates only. Annual rates were reconstructed from Number and same-release population, with original endpoint Rate checks. Types sum to total before displayed rounding.

3.3 Changing reference composition

For PCOS, Chinese rate changes contributed +271.51 per 100,000 to the changing contrast, reference-country rate changes −208.62, and reference-country share changes +88.65 (Table 3; Figure 3). The share component accounted for 58.5% of the +151.54 signed net change; the combined rate components accounted for +62.89. The corresponding share components were +88.11 ending in 2021 and +86.74 ending in 2022. Across country exclusions and the age-range sensitivity, the 2023 composition share ranged from 39.8% to 67.9%. A material reference-composition contribution persisted, although a majority contribution did not hold in every specification.

For primary PCOS infertility, the three components were +52.65, −38.09 and +31.84; for secondary infertility, they were +218.86, −170.53 and +56.81. Thus subtype and reference com-

position are intersecting dimensions of the same contrast change: the 69.4% subtype share and 58.5% composition share must not be added. Fixing all reference-country shares at 1990 instead yielded a 2023 total difference of −202.48, compared with −121.38 under current pooling. This fixed-share calculation follows one replacement path and therefore differs from the symmetric component.

Endometriosis's +2.73 net difference change combined Chinese rate changes of −10.39, reference-country rate changes of +13.50 and reference composition of −0.38. Its near-zero net change makes percentage contributions unstable; we report absolute components. Age-by-type component tables for each endpoint accompany Additional file 2.

Table 3: Symmetric decomposition of changes in standardized differences, 1990–2023.

Cause	Type	Net change	China rates	Ref. rates	Ref. shares
Endometriosis	Primary	+6.79	+0.02	+6.69	+0.08
Endometriosis	Secondary	-4.06	-10.41	+6.81	-0.46
Endometriosis	Total	+2.73	-10.39	+13.50	-0.38
PCOS	Primary	+46.40	+52.65	-38.09	+31.84
PCOS	Secondary	+105.14	+218.86	-170.53	+56.81
PCOS	Total	+151.54	+271.51	-208.62	+88.65

All components use per 100,000 females and sum to net change. Reference shares are country population shares within each age. Positive values move China minus reference upward. Percentages of a signed net change are accounting summaries, not causal fractions.

3.4 Age patterns and secondary analyses

Ages 35–39 and 40–44 contributed +31.07 and +33.60 per 100,000, respectively, to the PCOS difference change, together 42.7% of the total. These are standard-weighted contrast contributions, not case counts. The age-by-type table separates Chinese rates, reference rates and reference composition within each cell.

China's PCOS-related infertility count rose by 783,425. A separate case-count decomposition assigned +860,721 to age-specific prevalence, offset by $-68,426$ from female population size and $-8,869$ from age distribution within ages 15–49. This is consistent with increasing modeled prevalence despite demographic offsets; it does not identify its causes. The total PCOS difference remained negative across reference exclusions. The endometriosis total difference changed sign across country exclusions. Rounding the published standard weights within their

evaluated precision did not reverse the main 2023 signs. Secondary YLD, trend and benchmark results are retained in Additional file 1. Forecast candidates failed the comparative validation rule, so no long-range forecasts are reported.

4 Discussion

The principal finding is that subtype composition and reference population composition both affect the interpretation of international infertility contrasts. China's negative PCOS-related infertility difference narrowed, but a substantial part of that change corresponded to population redistribution among reference countries. At the same time, secondary infertility accounted for most of the total contrast change, while a primary-infertility difference persisted. These results concern how an international comparison is constructed and what its summary conceals.

This is an incremental contribution to an extensively studied field. Previous work already examined these causes, their primary and secondary infertility attribution, China/global patterns and standard trend and decomposition methods [4, 5, 6, 7]. Our contribution is the joint accounting of subtype and within-age reference-country shares, with sensitivity to country inclusion and endpoint. We do not propose a new decomposition method or claim that GBD 2023 or subtype stratification alone is novel. The value is an interpretable application that changes the substantive reading of apparent convergence.

Age standardization removes changes in the chosen age weights, but it does not fix the mix of countries within an age group. When countries with different infertility rates gain or lose population share, a pooled reference can move even with unchanged country rates. The symmetric identity quantifies this component separately from the rate component and avoids assigning the entire rate–share interaction to one replacement order. The fixed-1990-share sensitivity gives a complementary contrast under a different reference definition. Neither is a preferred target rate or a causal counterfactual.

The subtype results further discourage describing narrowing as improved reproductive health. Both Chinese and reference PCOS rates increased. Secondary infertility's larger contribution does not show that individual women moved from primary to secondary infertility, nor that parity history caused the increase. The GBD categories reflect modeled estimates and survey definitions; diagnosis, fertility intentions, exposure to pregnancy, treatment and underlying dis-

ease attribution could all affect interpretation, but were not directly measured. The all-female denominator cannot estimate the risk among couples trying to conceive.

For endometriosis, the final-year near cancellation should be viewed as an unstable contrast. Independent download checks reproduced the source values, yet these checks cannot determine whether the abrupt terminal-year change reflects new evidence, model behavior or an actual epidemiological shift. Analyses ending in 2021 or 2022 gave a materially different overall picture. This unresolved source-level issue warrants investigation with independent evidence before a substantive convergence claim. Retaining the earlier endpoints and full annual trajectories makes the limitation visible.

Case-count accounting answers a different public-health question. The PCOS increase occurred despite offsets from China's reproductive-age female population size and age composition. The age-specific prevalence component includes any changes embedded in the modeled rates; calling it an epidemiological component does not isolate biological risk, diagnosis or policy. Clinical guidelines support individualized reproductive and broader health assessment [1, 2]; these ecological comparisons cannot select interventions, estimate treatment benefits or demonstrate service inequity. They can identify patterns that surveillance and independent studies should examine.

The most serious limitation is unavailable joint uncertainty draws. We cannot propagate GBD uncertainty through pooling, standardization, subtype shares or the new decomposition. A 58.5% or 69.4% point estimate should therefore not be read as a precisely established population parameter. Country exclusions test dependence on the reference; they cannot replace draw-based intervals. Bootstrap resampling of the deliberately selected G20 countries would answer another question and would not recover GBD uncertainty. Conditional trend intervals are confined to the supplement and do not establish cross-country differences.

Other limitations include sparse or heterogeneous upstream evidence, modeled cause allocation, the distinction between impairment and disease prevalence, the economically selected comparison group, and the absence of external epidemiological validation. Standard weights were available only as a rounded published reproduction. Several analyses were added after inspection, including endpoint checks and the reference-composition decomposition; their exploratory status requires transparent reporting and future corroboration. No release-to-release comparison was performed, so differences from previous publications cannot be assigned to

real changes rather than GBD revision. These constraints limit the strength of inference even after formatting and numerical checks.

5 Conclusions

Within GBD 2023 point estimates, China's apparent convergence toward other G20 countries for PCOS-related infertility partly reflected changing reference-country population shares and concealed contrasting primary and secondary infertility patterns. Endometriosis near parity in 2023 depended strongly on the terminal year. International summaries should distinguish subtype, age and reference composition; these results require propagated uncertainty and independent corroboration before causal or policy conclusions.

List of abbreviations

AAPC, average annual percentage change; ARIMA, autoregressive integrated moving average; ASPR, age-standardized prevalence rate restricted here to ages 15–49; GATHER, Guidelines for Accurate and Transparent Health Estimates Reporting; GBD, Global Burden of Disease; G20, Group of Twenty; IHME, Institute for Health Metrics and Evaluation; MAE, mean absolute error; OLS, ordinary least squares; PCOS, polycystic ovarian syndrome; YLD, years lived with disability.

Declarations

Ethics approval and consent to participate

This study analyzed aggregated GBD estimates without access to identifiable individual records or recruitment of participants. The responsible authors must document the applicable institutional determination before submission. No institutional ethics approval or waiver has been invented for this draft.

Consent for publication

Not applicable: the manuscript contains no individual person's data, images or identifiable details.

Availability of data and materials

The derived datasets supporting the reported results and their analysis code are provided in Additional file 2 and the public GitHub repository [9]. Original GBD inputs are obtained from IHME through the Results Tool [8], subject to its data-use terms and access requirements. Exact dimensions, identifiers, download date, hashes and acquisition specifications are supplied. The original CSVs are retained in the research archive; they are not included in the public additional file. Re-running the full source-to-result pipeline requires obtaining these IHME inputs. The repository contains version history; the submission manifest records the analysis revision. A permanent data/code archive identifier remains to be finalized before submission.

Competing interests

Pending verified declarations from all human authors. An absence of competing interests has not been assumed.

Funding

Pending author-provided funding sources, grant numbers and funder roles. No funding status has been assumed.

Authors' contributions

Pending the actual human author list and verified individual contributions. All-author approval remains to be documented.

Acknowledgements

We acknowledge the GBD data resource and the contributions of its data providers and collaborators. Any additional named acknowledgements and required permissions remain for the

human authors to verify.

Additional files

Additional file 1 (PDF, with editable TeX source). Supplementary methods, secondary case-count and trend analyses, endpoint checks, forecast validation, acquisition audit and GATHER reporting map; supplementary tables and figures.

Additional file 2 (ZIP). Machine-readable derived result tables, executable Python source, configuration, dependencies, acquisition and validation manifests, and a README specifying licensed IHME inputs needed to reproduce the analysis. No synthetic research observations or raw licensed downloads are included.

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Figure legends

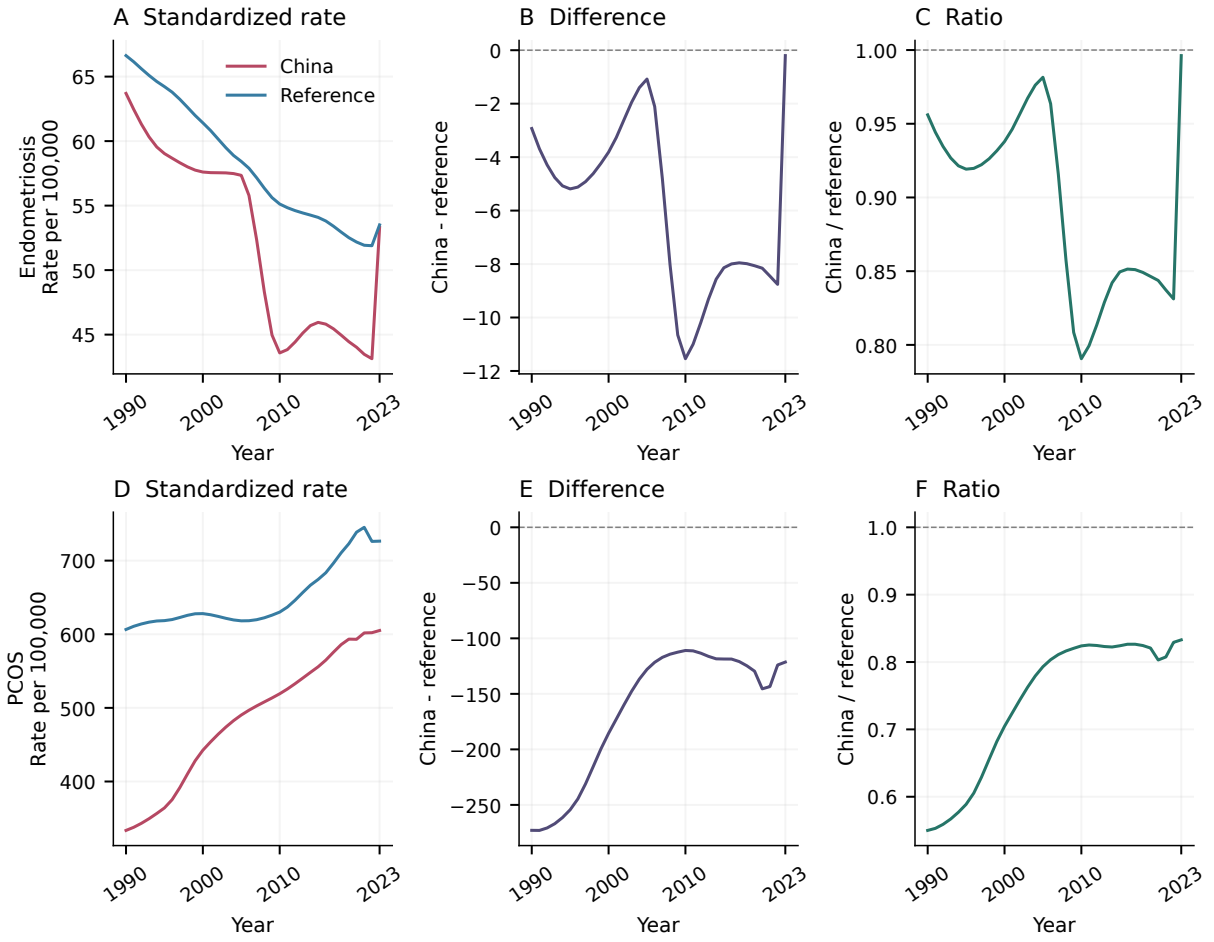


Figure 1: Annual total infertility rates and China–reference contrasts. Rates are standardized within ages 15–49. The reference pools population and modeled cases within age across 18 countries. Point trajectories have no propagated GBD uncertainty. The terminal-year endometriosis change is a source-estimate feature requiring independent corroboration.

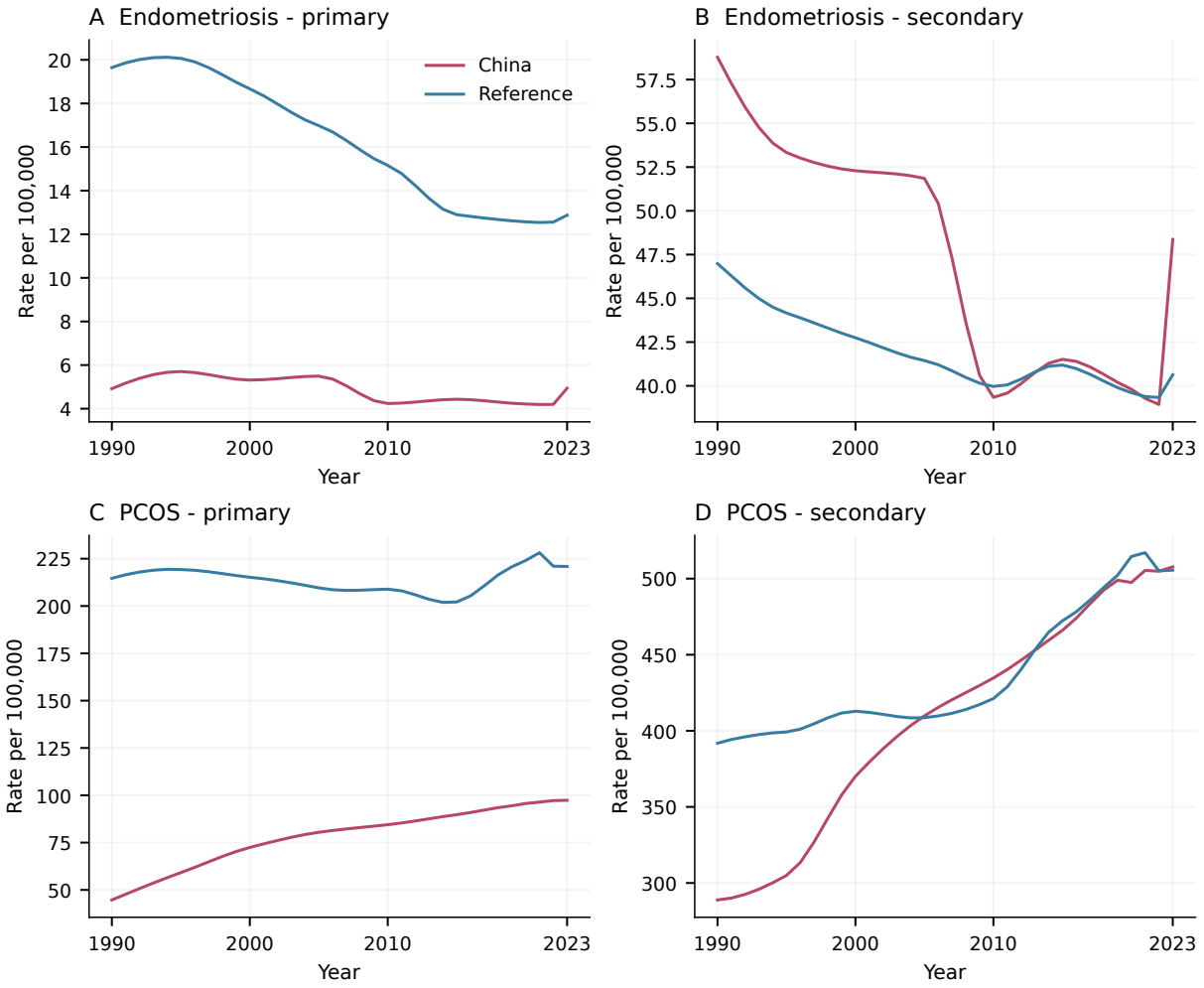


Figure 2: Primary and secondary infertility trajectories. The same restricted-age standard and annual population-pooled reference are used for both types. Opposing subtype contrasts can cancel in the total. These modeled categories do not track transitions in individual women.

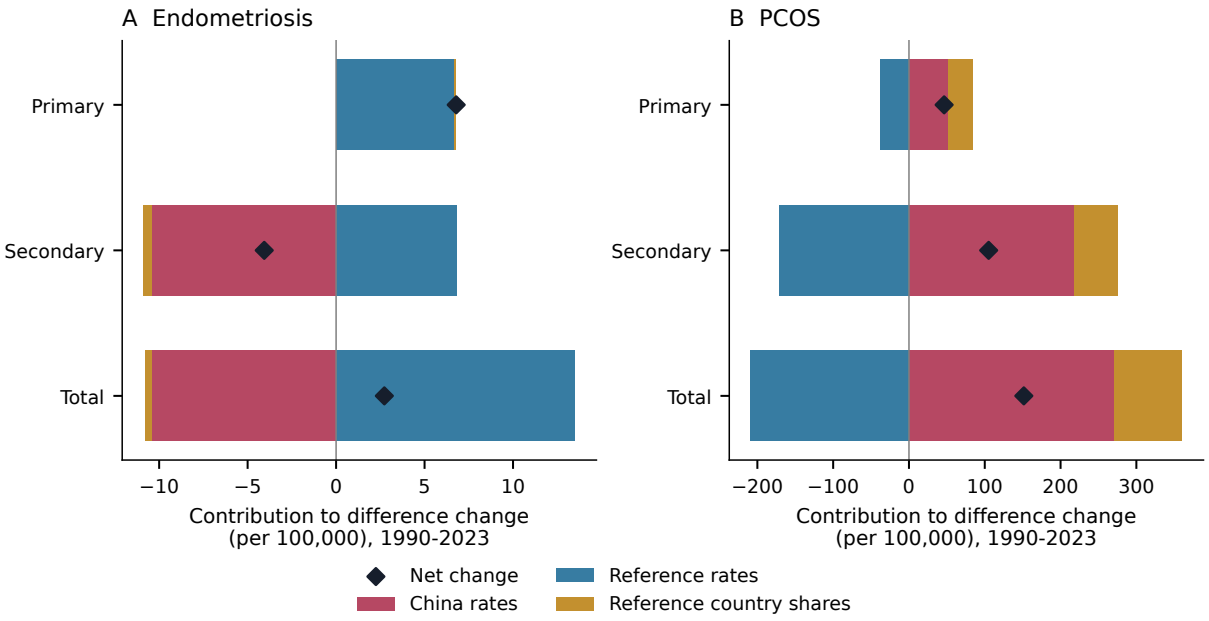


Figure 3: Rate and reference-composition contributions to contrast changes. Contributions sum to the 1990–2023 change in China minus reference, separately by infertility type. Diamonds denote net changes. Positive components move the signed contrast upward. Reference-country shares vary within age; fixed standard age weights are used throughout. The decomposition was added during manuscript review and is descriptive.